

Contactless Sleep Sensor

Communication Protocol

V1.15

Apr 21th 2025

HISTORY

Ref	Date	Change Reason
v1.0	2022/6/30	Creation
v1.1	2022/8/13	Add Device UUID
v1.2	2022/11/17	Add type and cmd description
v1.3	2023/7/7	Description for presence detection
v1.4	2023/8/24	Unified description
v1.5	2023/8/31	Unified description
v1.6	2023/9/1	Unified description
v1.7	2023/9/13	Add set in bed thresh
v1.9	2023/11/21	Add get in bed thresh
v1.12	2023/3/25	Add running time body energy
v1.13	2024/4/29	0x03e8 Add Body move energy and range
v1.14	2024/11/19	Add description
V1.15	2025/4/21	Add 0x03ed: In bed learning threshold

Name	magic (1 byte)	ver (1 byte)	type (1 byte)	cmd (1 byte)	req id (int)	timeout (short)	content len (int)	Func tag (2 bytes)	Data	↑radar to host ↓ host to radar	Remarks	example hex
Radar ID	0x13	0x01	response 0x00, request 0x01, request one way 0x02	Heartbeat 0x00, Request 0x01, Reponse 0x02	auto increment from 1	(0-32768)	6	0x0001	XX XX XX XX	↓		13 01 01 01 00 00 00 23 00 0A 00 00 00 06 00 01 00 00 00 00
							20		radar type(1 byte) + hw version (4 bytes) + Unique Device ID (13 bytes)	↑	Radar type (breath\heart rate 0x01) + Hardware version+ Unique Device ID. Upload every second until a response is received and the connection is considered successful.	13 01 01 01 00 00 00 23 00 0A 00 00 00 14 00 01 01 02 01 4A 00 13 36 01 11 50 48 57 35 33 38 04 72 20
Vital Data							38	0x03e8	32 bytes in total Breath BPM float Breath curve float Heart rate BPM float Heart rate curve float Target distance float Signal strength of the target (for presence detection) float Valid bit ID int Body Move Energy float Body Move Range float	↑	Radar reports data to the host. Valid tag: 0: invalid breath/heart rate 1: valid breath 2: valid breath and heart rate For presence detection, if the signal strength is 0, there is nobody in bed.	13 01 02 01 00 00 00 BB 00 0A 00 00 00 1E 03 E8 00 00 00 00 B9 83 B1 00 00 00 00 00 BB C3 65 C2 3F 19 99 9A 41 EC 96 C4 00 00 00 00
Report Interval							6	0x03e9	XX XX XX XX (Int, big endian)	↓	ReportInterval. For example, if it is set at 20, the breath/heart rate is reported every 1 second. Set value range 1~60000 (frame repeat time)	

									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03ea	0x00 0x00 0x00 0x00	↓	Command to read the ReportInterval (unit: 50ms) .	
									XX XX XX XX (Int, big endian)	↑	To Read the ReportInterval (unit: 50ms). For example, if the value is 20, the breath/heart rate is reported every 1 second.	
Working Distance							6	0x03eb	XX XX XX XX (Float, big endian)	↓	To set the Working Distance for the radar to search for the target (Valid: 0.6~3.0m. Default setting: 1.5. Unit: m). If it is set at 1.2, the radar will consider signals within a range of 0.5~1.2m as the chest of the user. 0~3.0 (m)	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03ec	0x00 0x00 0x00 0x00	↓	Command to read the Working Distance	
									XX XX XX XX (Float, big endian)	↑	0: Automati search without limit. . >0: the max. Working Distance(unit:m).	
In bed learning threshold							6	0x03ed	0x00 0x00 0x00 0x00	↓	To start learning in bed threshold, Please make sure there is no one near the radar before sending this command.	

									XX XX XX XX (Float, big endian)	↑	In bed learning	
No Breath Alert Timer							6	0x03ee	XX XX XX XX (Int, big endian)	↓	To set the NoBreathAlert threshold (unit: s). 0: turn off the function. 30~86400: the time threshold to trigger the NoBreathAlert. 30~86400 (second) or 0	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03ef	0x00 0x00 0x00 0x00	↓	Command to read the NoBreathAlert threshold (unit: s).	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. 30~86400: the time threshold to trigger the NoBreathAlert.	
							6	0x03f0	0x00 0x00 0x00 0x00	↑	NoBreathAlert	
Low Breath Bpm Alert							6	0x03f1	XX XX XX XX (Int, big endian)	↓	To set the LowBreath(BPM)Alert threshold (unit: s). 0: turn off the function. N: the LowBreath(BPM)Alert threshold. 0~1024 (BPM)	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03f2	0x00 0x00 0x00 0x00	↓	Command to read the LowBreath(BPM)Alert threshold (unit: s).	

									XX XX XX XX (Int, big endian)	↑	0: turn off the function. N: the respiratory BPM threshold.	
							6	0x03f3	0x00 0x00 0x00 0x00	↑	LowBreath(BPM)Alert	
High Breath Bpm Alert							6	0x03f4	XX XX XX XX (Int, big endian)	↓	To set the HighBreath(BPM)Alert threshold (unit: s). 0: turn off the function. N: the HighBreath(BPM)Alert threshold. 0~1024 (BPM)	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03f5	0x00 0x00 0x00 0x00	↓	Command to read the HighBreath(BPM)Alert threshold (unit: s).	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. N: the respiratory BPM threshold.	
							6	0x03f6	0x00 0x00 0x00 0x00	↑	HighBreath(BPM)Alert	
Time to trigger Low/Hight Breath Bpm Alert							6	0x03f7	XX XX XX XX (Int, big endian)	↓	To set the time interval to trigger the High/Low Breath(BPM) Alert (unit: s). Please note that the High/Low Breath(BPM) Alert will only be triggered when the detected respiratory rate is consistently higher/lower than the set value. 30~86400: the time interval to trigger the High/Low Breath(BPM) Alert (default	

											setting: 120). 30~86400 (second) or 0	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03f8	0x00 0x00 0x00 0x00	↓	Command to read the time interval to trigger the High/Low Breath(BPM) Alert (unit: s).	
									XX XX XX XX (Int, big endian)	↑	The time interval to trigger the High/Low Breath(BPM) Alert (unit: s).	
No Heart Alert Timer							6	0x03f9	XX XX XX XX (Int, big endian)	↓	To set the time threshold to trigger the the NoHeartAlert (unit:s). 0: turn off the function. 30~86400: the time threshold to trigger the NoHeartAlert. 30~86400 (second) or 0	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03fa	0x00 0x00 0x00 0x00	↓	Command to read the time threshold to trigger the NoHeartAlert.	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. 30~86400: the time threshold to trigger the NoHeartAlert.	

							6	0x03fb	0x00 0x00 0x00 0x00	↑	NoHeartAlert	
Low Heart Bpm Alert							6	0x03fc	XX XX XX XX (Int, big endian)	↓	To set the LowHeartAlert (BPM) threshold (unit: s). 0: turn off the function. N: the LowHeartAlert (BPM) Alert threshold. 0~1024 (BPM)	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x03fd	0x00 0x00 0x00 0x00	↓	Command to read the LowHeartAlert (BPM) threshold (unit: s).	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. N: the LowHeartAlert (BPM) threshold.	
							6	0x03fe	0x00 0x00 0x00 0x00	↑	LowHeartAlert (BPM)	
High Heart Bpm Alert							6	0x03ff	XX XX XX XX (Int, big endian)	↓	To set the HighHeart(BPM)Alert threshold (unit: s). 0: turn off the function. N: the HighHeart(BPM)Alert threshold. 0~1024 (BPM)	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x0400	0x00 0x00 0x00 0x00	↓	Command to read the HighHeart(BPM)Alert threshold (unit: s).	

									XX XX XX XX (Int, big endian)	↑	0: turn off the function. N: the HighHeart(BPM)Alert threshold.	
							6	0x0401	0x00 0x00 0x00 0x00	↑	HighHeart(BPM)Alert	
Time to trigger Low/High Heart Bpm Alert							6	0x0402	XX XX XX XX (Int, big endian)	↓	High/LowHeartAlert Timer. To set the time interval to trigger the High/Low Heart (BPM) Alert (unit: s). Please note that the High/Low Heart (BPM) Alert will only be triggered when the detected heart rate is consistently higher/lower than the set value. 30~86400: the time interval to trigger the High/Low Heart (BPM) Alert (default setting: 120). 30~86400 (second) or 0	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x0403	0x00 0x00 0x00 0x00	↓	Command to read the time interval to trigger the High/LowHeartAlert Timer (BPM) Alert (unit: s).	
									XX XX XX XX (Int, big endian)	↑	High/LowHeartAlert Timer (unit: s).	
Bed exit alert timer							6	0x0404	XX XX XX XX (Int, big endian)	↓	To set the time threshold to trigger the Bed Exit Alert (unit:s). 0: turn off the function. 1~N: the time threshold to trigger the Bed Exit Alert. 30~86400 (second) or 0	

									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed	
							6	0x0405	0x00 0x00 0x00 0x00	↓	Command to read the time threshold to trigger the Bed Exit Alert (unit:s).	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. 1~N: the time threshold to trigger the Bed Exit Alert	
							6	0x0406	0x00 0x00 0x00 0x00	↑	Bed Exit Alert	
No-Turn-Over alert timer							6	0x0407	XX XX XX XX (Int, big endian)	↓	To set the time threshold to trigger the No-Turn-Over Alert (unit: s). 0: turn off the function. 1~N: the time threshold to trigger the No-Turn-Over Alert. 30~86400 (second) or 0	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x0408	0x00 0x00 0x00 0x00	↓	Command to read the time threshold to trigger the No-Turn-Over Alert	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. 1~N: the time threshold to trigger the No-Turn-Over Alert	
							6	0x0409	0x00 0x00 0x00 0x00	↑	No-Turn-Over Alert	

Turn on/off Turn-Over Alert							6	0x040a	XX XX XX XX (Int, big endian)	↓	To turn on the Turn-Over Alert. 0: turn off the function. 1: algorithm based on users turning over in bed.	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x040b	0x00 0x00 0x00 0x00	↓	Command to read the status of the Turn-Over Alert	
									XX XX XX XX (Int, big endian)	↑	0: turn off the function. 1: algorithm based on users turning over in bed.	
							6	0x040c	0x00 0x00 0x00 0x00/0x01	↑	Turn-Over Alert	
Turn on/off body movement							6	0x040d	XX XX XX XX (Int, big endian)	↓	Turn ON/OFF counter for body movement. 0: OFF 1: ON	
									0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
							6	0x040e	0x00 0x00 0x00 0x00	↓	Command to read the body movement counter.	

									XX XX XX XX (Int, big endian)	↑	0: OFF 1: ON	
							6	0x040f	XX XX XX XX (float)	↑	The data value is the accumulated energy for every body movement.	
Get Radar ID							6	0x0410	0x00 0x00 0x00 0x00	↓	Get Radar ID	
							15		13 bytes Radar ID	↑	Radar ID	
Reset Radar							6	0x0411	0x00 0x00 0x00 0x00	↓	Command to reset the radar.	
							6		0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
Set In-bed detect thresh							6	0x0418	XX XX XX XX (Float, big endian)	↓	Command to set in bed thresh	
							6		0x00 0x00 0x00 0x01/0x00	↑	Successful/Failed.	
Get in-bed detect thresh							6	0x0419	0x00 0x00 0x00 0x00	↓	Command to get in bed thresh	

							6		XX XX XX XX (Int, big endian)	↑	Return in bed thresh	
Body move energy							6	0x041A	XX XX XX XX (Int, big endian)	↑	Running-Time body move energy (firmware version >=2.1.81)	
Upgrade steps: 1. Send the upgrade package and the radar responds; 2. Send the firmware in segments; 3.Complete the upgrade.												
Start Upgrade							6	0x0021	Size of the firmware, 524292 bytes for example	↓	Notice to upgrade and to send the size of the firmware.	
									0x00 0x00 0x00 0x01/0x00	↑	Radar responds: Successful/Failed.	
Firmware file							Data length	0x0022	Firmware + crc	↓	CRC16 verified for every frame. (Only the firmware content, without the header, length and the func tag) Max allowance for the content: 240bytes.	
									0x00 0x00 0x00 0x01/0x00	↑	Responds from the radar of whether the frame is successful or not.	
Finish upgrade							6	0x0023	0x00 0x00 0x00 0x00	↓	Succeded in sending the upgrade.	
									0x00 0x00 0x00 0x01/0x00	↑	Radar responds: Successful/Failed.	